

Supplements M1-M13 are the **Supplementary Methods**: they carry the methodological detail condensed out of the main text. Every statement in them is a fuller form of something the Methods section states or points at, and nothing reported in the main text depends on material that appears only there. Supplements S1-S9 are supporting analyses and extended results.

Supplement M1. Source data, setting, and sampling frame

The study used a single de-identified extract from the Epic EHR of Saint Francis Health System. Seven files were delivered: person, encounter, diagnosis, medication, procedure, and laboratory/flowsheet tables together with a medication-to-RxNorm mapping table. The clinical tables originated from the Caboodle enterprise data warehouse. The RxNorm mapping originated from the Clarity reporting database. Body mass index and systolic and diastolic blood pressure were obtained from the laboratory/flowsheet table. Each table was delivered as a flat file after patient and encounter keys had been replaced by one-way MD5 hashes, and was transferred by secure file transfer protocol. No names, medical record numbers, direct identifiers, or dates of birth were included.

The delivering data team restricted the person table to one administrative division of the health system and to patients who were current, valid, non-test, non-historical, not recorded as deceased, and aged 18-110 years at extraction. Eligible source encounters were completed on or before the extraction date, had at least one associated diagnosis, and had an inpatient, outpatient, observation, or emergency patient class.

Investigators prespecified the diagnosis code lists. Depression comprised ICD-9 codes 296.2, 296.3, 300.4, and 311 or ICD-10 codes F32.*, F33.*, and F34.1. Bipolar disorder comprised ICD-9 codes 296.0 and 296.4-296.8 or ICD-10 codes F30.* and F31.*. Schizophrenia-spectrum disorders comprised ICD-9 codes 295.* and 298.* or ICD-10 codes F20.*, F23.*, F25.*, F28.*, and F29.*.

The extract was assembled as a case-enriched sample. All patients with a depression diagnosis on the problem list were retained, and a random sample of patients without that flag was added at an approximate 4:1 unflagged-to-flagged ratio. The remaining tables were then linked to the person table. The delivered extract included 501,718 patients: 100,420 (20.0%) with and 401,298 (80.0%) without a problem-list depression flag.

Sampling used the problem list, whereas study eligibility used diagnoses recorded at individual encounters. A depression code entered at a visit is a documented diagnosis whether or not it was ever added to the problem list. In routine care the problem list is frequently not updated, so the two sets overlap without either containing the other. Of the 42,579 analysis-cohort patients, 12,530 (29.4%) qualified through an encounter-level depression diagnosis despite lacking the problem-list flag and therefore entered through the randomly sampled group. The analysis cohort is consequently a sample, not an enumeration, of the health system's patients with depression, and cohort TRD and subgroup proportions are not population prevalence estimates. The paired comparison between representations remains internally valid because both were evaluated in the same patients on the same held-out split.

The setting is a single community health system in Tulsa, Oklahoma, serving inpatient, outpatient, observation, and emergency care. Patients enter the cohort through routine antidepressant prescribing across those settings rather than through psychiatric specialty referral, and the index prescription may be written in any of them.

Supplement M2. Eligibility cascade, index selection, and temporal design

Eligibility was applied hierarchically from the full delivered extract. Patients were required to have all six of the following. 1. A qualifying MDD diagnosis. 2. No bipolar-disorder diagnosis and no schizophrenia-spectrum diagnosis, so that the cohort is unipolar depression. 3. An eligible antidepressant index prescription. 4. An MDD diagnosis recorded on or before the index date. 5. At least 730 days of pre-index history. 6. At least 365 days of post-index follow-up. Patients failing the chronological or observation prerequisites were assigned explicit rejection reasons for attrition accounting.

Table M1. Participant flow through the hierarchical eligibility filters. Each row applies one additional filter to the survivors of the row above. The first row is the delivered extract, a case-enriched sample by design (Supplement M1).

Stage	Remaining	Rejected at stage
Delivered extract (4:1 case-enriched)	501,718	—
MDD-diagnosed	144,110	357,608
Not bipolar AND not schizophrenia-spectrum	124,188	19,922

Stage	Remaining	Rejected at stage
Has antidepressant index prescription	107,636	16,552
Has MDD before index	103,458	4,178
≥2 years pre-index history	54,215	49,243
≥1 year post-index follow-up (final cohort)	42,579	11,636

Upstream data preparation assembled, for each patient, all antidepressant medication orders beginning on or after the date of the first documented depression diagnosis. The earliest start date in that set defined the index date, one per patient. Among candidate index orders, 57.9% begin on the same date as the first recorded depression diagnosis.

No post-index property was used to qualify the index exposure. Dose adequacy, exposure duration, subsequent response, and treatment changes were not considered, so the prediction point is the moment the prescription is written. Post-index data entered only through the requirement for at least 365 days of follow-up and outcome ascertainment during those 365 days. Predictors used only information within the 730 days ending on the index date. Although recorded history extends further back for many patients (median 1,792 days), clinical content outside the fixed lookback window was not read. Total recorded history length was itself retained as a predictor.

Antidepressant exposure before the index date did not exclude a patient. The index is tied to the first documented depression diagnosis rather than to the first antidepressant exposure. Consequently 24.0% of the cohort had some recorded pre-index antidepressant exposure, and 14.7% had at least one pre-index course meeting the 42-day threshold used for the adequate-trial predictors. The index is therefore best described as the first antidepressant prescription after a documented depression diagnosis, and not as the first antidepressant treatment or first adequate trial.

Supplement M3. Outcome construction and inferential boundary

The binary outcome was generated in the upstream R data-preparation pipeline independently of the predictors and supplied to the modeling workflow as a fixed label. TRD was assigned when a patient with MDD received at least three distinct antidepressant treatments within 365 days of the index date, the index agent

counting as the first. Patients with fewer than three treatments were classified TRD-negative.

A treatment is a distinct antidepressant agent rather than an order or prescribing event. Addition of a second agent while the current antidepressant continues is augmentation and does not advance the outcome count. Restarting a previously used agent also does not advance it. Augmentation is available as a pre-index predictor where its definition is met.

The 365-day follow-up requirement ensures that every patient has the observation window needed to determine the label, so a negative label reflects an absence of further antidepressant treatments rather than an absence of observation. It does not guarantee that care delivered outside the health system was captured.

The label approximates, but does not establish, the consensus characterization of TRD as inadequate response to at least two antidepressant trials of adequate dose and duration [4,5]. Non-response is inferred from switching rather than measured, because standardized symptom severity is not systematically recorded in this extract. Dose and duration adequacy were not verified for the counted post-index treatments. Changes attributable to intolerance, adverse effects, cost, formulary restriction, fragmented care, patient preference, or prescriber practice are therefore counted in the same way as changes attributable to non-response. Switching also depends on continuity, access, insurance, and clinical practice, which may differ across demographic groups [16,17], so the target can encode health-care process and inequity in addition to pharmacologic non-response. The label was not validated against chart review or a symptom-severity criterion. Future validation should compare it with adjudicated treatment histories and symptom trajectories, and should test a higher-specificity outcome requiring documented adequate courses.

Supplement M4. Predictor selection and the typed feature representation

The candidate predictor set was specified before model fitting and before predictor-outcome associations were examined. The approach followed manual clinical and literature-based selection, as in the treatment-resistance model of Perlis [18], rather than univariate screening or automated selection. No variable was retained or removed because of its observed association with TRD, and no automated selection step ran at any stage.

The selected domains reflect prior treatment-resistance and antidepressant-response research: depression severity and recurrence; psychiatric and substance-use comorbidity, medical comorbidity, prior antidepressant exposure and medication burden, health-care utilization, and sociodemographic characteristics [18,19]. Unlike clinical-trial studies that use scheduled symptom scales and structured interviews, this study is restricted to structured fields recorded in routine EHR care, so depression severity enters as diagnostic coding rather than as a rating-scale score. Prior EHR prediction studies informed the feasible domains [20,21].

Prescribing-constraint flags were retained because they describe whether escalation options may be limited. Sociodemographic and social-determinant fields were retained so that their direct contribution could be tested by the semantic-feature ablation rather than rendered unobservable by design. The only prespecified fields later removed were body mass index and systolic and diastolic blood pressure, for the missingness reasons in Supplement M6 rather than because of any association with the outcome.

FEATURE assigns an explicit type to every field. Quantitative counts and durations are continuous, including encounter count, pre-index history length, age, per-agent adequate-trial counts, and medication burden. Single binary flags and multi-label indicator sets are boolean, including psychiatric and medical comorbidity, prescribing constraints, substance-use categories, and social-determinant categories. Single-valued nominal variables are categorical, including sex, preferred language, marital status, religion, smoking status, race/ethnicity, MDD recurrence, and coded MDD severity. An adequate prior antidepressant trial requires at least 42 days of continuous exposure to an agent. The complete predictor inventory, with each field's type and encoded form, is Multimedia Appendix 1.

Supplement M5. Deterministic narrative and embedded representation

The same pre-index record was rendered deterministically into a human-readable Markdown narrative, and two examples are reproduced in Supplement S4. Rendering is rule-driven and reproducible: no generative language model participates in construction. This choice prioritizes traceability over fluency, because an unsupported generated statement would silently alter the patient's model input [22].

What the renderer is given, and what it is not. It walks a fixed template of section headings and field labels, in a fixed order, over the field inventory selected in Supplement M4. That is the same inventory the feature vector receives. Absent findings are printed explicitly and unrecorded ones as a literal token, so every patient reaches the encoder through the same constant form and the narrative’s length varies only with how much of that inventory is populated. The renderer never reads the record outside the 730-day lookback window, and it never reads a field that predictor selection did not choose. An untailored encoding of a whole record is therefore a different experiment and not a variant of this one. The main text states why it was not attempted (Methods, *Predictors and patient representations*).

Each narrative was mapped to a fixed-length vector with a pre-trained sentence-transformer encoder [28]. The four independently evaluated encoders were bge-small-en-v1.5 [23], bge-en-icl [24], Qwen3-Embedding-4B, and Qwen3-Embedding-8B [25]. This serialization-and-encoding strategy follows evidence that general-purpose language-model embeddings of serialized EHR records can perform competitively with purpose-built EHR foundation models across prediction tasks [15].

FEATURE and EMBEDDED use the same timeline-sliced source record but not an identical set of rendered fields. The field-level crosswalk in Supplement S8 gives, for every source field, its FEATURE encoding, narrative rendering, missing-value rule, and temporal availability. Because the field inventories differ, the comparison is between complete representation pipelines and is not a controlled comparison of numeric versus language encoding of identical information.

Supplement M6. Missing-data handling

No statistical imputation was performed. Mean body mass index and mean systolic and diastolic blood pressure were present in the intermediate feature file but removed from the FEATURE representation at load time. Each value is the within-patient mean across that patient’s own pre-index encounters, not a mean across patients. Mean body mass index was missing for 22.1% of patients, and at least one vital sign was absent for 21.0%. Body mass index was missing for 28.8% of TRD-positive and 20.6% of TRD-negative patients, an approximately 8-percentage-point difference.

That association rules out missing completely at random but does not establish missing not at random, which cannot be identified from observed data: missingness may depend on health-system contact that

the recorded predictors capture only partly. The variables were removed because no defensible imputation model was available from the observed predictors alone. No missingness indicators were created for the continuous block, and no NaN values reached an estimator. Their removal is stated in the main text, Methods, *Missing data and sample partition*.

Missing categorical values were encoded as a separate level by the FEATURE one-hot encoder and as the literal token “Missing” in EMBEDDED. Marital and smoking status are nearly complete (<0.5% missing). Religion is missing for 29.4% overall and shows a monotone age gradient, from 43.8% at ages 18-29 to 17.5% at ages 65 or older, so the missing-category level itself may carry age-related information. Religion was retained with that indicator level, so the models can use the fact of an unrecorded value as well as the recorded ones.

Supplement M7. Sample size and events per variable

No formal a-priori sample-size calculation was performed, because cohort size was fixed by the eligibility cascade. Model dimensionality is summarized as events per variable (EPV), the number of TRD-positive patients in the training set divided by the number of encoded input columns. The numerator is 5,964 events rather than the total training sample of 34,063, because what limits how precisely a coefficient can be estimated is the size of the minority class. The denominator is the matrix dimension actually supplied to the estimator.

FEATURE contains 92 encoded columns, giving $EPV = 5,964/92 \approx 64.8$. EMBEDDED dimensionality is encoder-specific: bge-small-en-v1.5 has 384 dimensions ($EPV \approx 15.5$), Qwen3-Embedding-4B has 2,560 ($EPV \approx 2.3$), and bge-en-icl and Qwen3-Embedding-8B each have 4,096 ($EPV \approx 1.5$). Only the smallest encoder exceeds the conventional threshold of 10 events per variable.

EPV was developed for lower-dimensional regression and is not a complete adequacy criterion for high-dimensional regularized learners. The three larger encoders are therefore interpreted as high-dimensional regularized models rather than as the low-dimensional regressions for which EPV was devised. The low ratios nevertheless emphasize overfitting risk. EPV does not account for the discrimination ranking: bge-small-en-v1.5 has the most favorable ratio and is the weakest encoder.

Supplement M8. Train/test split, comparability, and leakage safeguards

One stratified 80:20 train/test split was generated once and reused for all models, representations, encoders, and ablations. The training set includes 34,063 patients (5,964 TRD-positive, 17.5%) and the test set 8,516 (1,491 TRD-positive, 17.5%). Standardized mean differences were calculated for every predictor as the train-minus-test mean difference divided by the pooled standard deviation. Across 100 predictor rows the largest absolute SMD is 0.036 and none reaches 0.10, and the distributions themselves are Supplement S6. The largest SMD arises from a rare social-determinant flag present in 22 training patients and no test patient. Outcome frequency is matched by stratification rather than by observation.

All data-dependent preprocessing occurred inside the cross-validation pipeline. On each fold, numeric scaling parameters and categorical levels were learned only from that fold's training rows and then applied to its validation rows. The final pipeline was fitted on the full training set and evaluated once on the test set.

In the retrieval analyses all test identifiers were removed from the searchable index. Each test patient was a query anchor, but every neighbor and neighbor outcome came from the training set, which prevents self-retrieval, test-to-test outcome propagation, and any direct use of a test label in prediction. Train/test comparability supports internal evaluation but does not establish transportability, because both partitions inherit the same case-enriched sampling frame.

Supplement M9. Standard machine-learning pipeline

Four classifiers were fitted to each representation: logistic regression, random forest, gradient boosting, and XGBoost. For FEATURE, the column transformer standardized the numeric block, one-hot encoded categorical variables with binary categories collapsed and unknown inference-time levels ignored, and cast boolean indicators to integer without further transformation. EMBEDDED entered through the numeric branch only.

Each classifier and its preprocessing transformer were wrapped in a single scikit-learn pipeline. Hyperparameters were selected by five-fold grid search on the training set using ROC AUC as the optimization criterion. The best configuration was refitted on all training patients

and generated held-out probabilities. The standardizer and the one-hot encoder both estimate their parameters from data, so fitting either outside the cross-validation loop would let held-out rows inform their own transformation. Bundling them with the estimator prevents this. With no imputation anywhere in the pipeline, these are the only two fitted preprocessing steps, so the cross-validated estimates carry no preprocessing leakage. Tuning grids are keyed by classifier only and are identical across representations.

Supplement M10. Neighbor-weighted retrieval prediction

For EMBEDDED only, the predicted TRD probability for query patient q was the weighted mean of binary TRD labels among $K = 50$ retrieved training neighbors:

$$\hat{P}(\text{TRD} \mid q) = \frac{\sum_{i=1}^K w_i y_i}{\sum_{i=1}^K w_i},$$

where $y_i \in \{0, 1\}$ is neighbor i 's TRD label and w_i its non-negative weight. No model is fitted anywhere in this arm. The predictor is the observed outcome rate of a retrieved set of real patients. That is what makes it a direct test of the premise that a patient's course can be read from the courses of their closest analogues.

Two weighting strategies are reported. Under uniform weighting all weights are equal and the prediction is the raw TRD rate of the retrieved set. Under cosine weighting each neighbor is weighted by its cosine similarity to the anchor. Neighborhood concentration was summarized by the effective sample size,

$$\text{ESS} = \frac{\left(\sum_{i=1}^K w_i\right)^2}{\sum_{i=1}^K w_i^2},$$

which equals K under uniform weighting and falls as weight concentrates on fewer neighbors. Cosine weighting leaves it essentially unchanged, at a mean ESS of 49.98 against 50.00 under nearest retrieval. That is the mechanical reason the two weightings return

nearly identical discrimination. Within a nearest-neighbor set the similarities are all high and all close together, so the weights are nearly uniform whatever the formula.

Two retrieval schemes were evaluated. *Nearest* selects the K training patients of highest cosine similarity. *Random* draws K training patients uniformly, as a negative control. This arm is embedding-only, because cosine similarity is not well defined over mixed categorical and numeric features.

All test identifiers were removed from the searchable index. Each test patient was a query anchor, and every neighbor and neighbor outcome came from the training set, which prevents self-retrieval, test-to-test outcome propagation, and any direct use of a test label in prediction.

Supplement M11. Semantic-feature ablation specifications

The ablation measures direct-input reliance. For each target concept, patient values were reassigned through a random one-to-one cohort-wide donor permutation, which retains the same marginal set of values while severing the association between each patient's concept value and their outcome. Section-level ablations exchange a complete narrative section. Field-level ablations exchange only the specified field. Perturbed narratives were re-embedded, but the baseline classifiers were kept frozen, so the resulting change in ROC AUC answers whether the fitted embedding pipeline relies on that input, not whether a newly trained model could substitute correlated information. Like every other predictor, all six concepts are measured at or before the index date, and permuting one leaves the remainder of the narrative untouched.

Psychiatric history (section). Present/absent indicators for anxiety disorder, social anxiety disorder, obsessive-compulsive disorder, posttraumatic stress disorder, adjustment disorder, dysthymia, insomnia, and any substance-use disorder, any suicidal-ideation or suicide-attempt code in the two-year lookback, and named substance categories including alcohol, cannabis, cocaine, nicotine, opioid, and sedative/hypnotic disorders. It carries documented diagnoses rather than treatments.

Medication burden (section). The count and names of distinct active drug ingredients at the index date across all therapeutic classes, and the count and names of distinct non-steroidal anti-inflammatory

ingredients prescribed for at least seven days. It stands in for overall medical complexity and pill burden, not psychiatric treatment specifically.

Treatment exposure (section). Four things. Pre-index counts of adequate trials for selective serotonin reuptake inhibitors, serotonin-norepinephrine reuptake inhibitors, bupropion, mirtazapine, and vortioxetine, each requiring at least 42 continuous days on the agent. Total benzodiazepine days covered during the lookback. Hypnotic agents prescribed for at least seven days. And augmentation, defined as overlap of an antidepressant with lithium, an antipsychotic, or buspirone for at least 14 days. It is distinct from all-class medication burden and from the post-index agents used to construct the outcome.

Treatment contraindications (section). Seizure disorder, which weighs against bupropion, and uncontrolled hypertension, which weighs against serotonin-norepinephrine reuptake inhibitors. The concept is constraint on escalation options rather than general medical comorbidity, which the narrative reports in a separate section that was not permuted.

Race/ethnicity (field). The recorded race/ethnicity category, including an explicit unrecorded level, permuted within an otherwise unchanged sociodemographic section. Sex, age, preferred language, marital status, religion, and smoking status were not altered in this ablation.

Social determinants of health (field). ICD-10 Z-code categories recorded during the lookback: education/literacy, employment, occupational exposure, housing/economic circumstances, social environment, upbringing, primary support group/family circumstances, psychosocial circumstances, and legal/criminal circumstances. The narrative renders “None Recorded” when no qualifying code is present. Only this field was permuted.

The ablation does not determine whether outcome labeling or care processes are equitable, whether a permuted attribute can be reconstructed from correlated inputs, or whether discrimination and calibration differ across subgroups. A negligible race/ethnicity delta, for example, would not exclude the presence of sociodemographic information in utilization, diagnosis, or treatment patterns.

Supplement M12. Evaluation coverage

Standard machine learning, the cross-encoder comparison, and the semantic ablation all ran on the complete 42,579-patient

cohort. All three scored the shared 8,516-patient test set, with Qwen3-Embedding-8B as the primary encoder.

Retrieval used nearest and random schemes crossed with uniform and cosine weighting, on all four encoders. The retrieval scheme dominated the weighting strategy throughout: the two weightings were nearly indistinguishable under both schemes, while nearest and random separated cleanly (main text, Table 6). That is what establishes that the nearest-versus-random ordering and the shortfall against a trained classifier both generalize across encoders (main text, Table 7).

Supplement M13. Metrics and uncertainty

Discrimination was measured by ROC AUC. Calibration was summarized by intercept and slope, with ideal values of 0 and 1. The extract was case-enriched, so the average predicted risk is anchored to this cohort's own TRD frequency. Carrying the models to a population with a different rate would require rescaling the probabilities to it.

Continuous probabilities were dichotomized at the test-set threshold maximizing Youden's J, and sensitivity, specificity, and positive and negative likelihood ratios were calculated from the resulting confusion matrix. Selecting and evaluating the threshold in the same test set produces optimistic operating characteristics. These estimates characterize the ROC curve and are not proposed for deployment. A clinical threshold would have to be selected entirely within development data and evaluated in a separate temporal or external cohort.

Metric confidence intervals were estimated from 1,000 bootstrap resamples of the test-set patients, drawn with replacement. For ablation contrasts, the same patient resample was applied to the baseline and perturbed prediction vectors and the ROC AUC difference recomputed. The point estimate is the full-sample difference, and the 2.5th and 97.5th percentiles of the bootstrap distribution form the 95% confidence interval. Paired resampling preserves within-patient correlation and avoids overstating the uncertainty of a difference.

Representation contrasts used the same paired bootstrap. Classifier-matched comparisons held the learner constant and varied only FEATURE versus EMBEDDED. The best-feature-versus-best-embedded contrast compares each representation's strongest classifier, but that classifier was identified post hoc on the test set, so the contrast is interpreted descriptively. One resample matrix was reused across all contrasts to keep them mutually comparable.

No equivalence or noninferiority margin was prespecified. A paired confidence interval containing zero indicates that superiority was not established at the precision achieved. It does not establish equivalence.

Supplement S1. Effective dimensionality of the embedded representation

Individual embedding dimensions carry no clinical meaning. Per-feature interpretability, available for the feature vector in main-text Figure 6, therefore does not transfer to the 4,096-dimensional embedded representation. Instead we characterize *how many* latent dimensions carry the predictive signal, using three complementary views. All figures are for the primary Qwen3-Embedding-8B encoder on the held-out test set.

S1.1 Sparsity and cumulative built-in importance

The best logistic-regression fit used an elasticnet penalty ($l1_ratio$ 0.25, C 0.01) that zeroed most coefficients. Only 385 of the 4,096 dimensions carry nonzero weight, so the signal is concentrated in a sparse subset of them. We then ranked dimensions by each fitted classifier's native importance, using $|coef|$ for logistic regression and $feature_importances_$ for the tree ensembles, and accumulated the importance mass. That confirms the concentration. For logistic regression, 80% of the coefficient magnitude falls in roughly 179 of the 4,096 dimensions and 90% in roughly 236 (main-text Figure 7). TRD-relevant information is therefore carried by a modest subset of embedding dimensions rather than spread across the space.

S1.2 Cumulative univariate correlation

An importance ranking is model-specific. As a model-agnostic check we ranked dimensions by the absolute Spearman correlation between each dimension and the outcome, $|\rho(dim, y)|$, and overlaid that baseline against per-classifier curves ranking dimensions by $|\rho(dim, risk\ score)|$ (Figure S1). Divergence between the model-agnostic baseline and a classifier's curve would flag dimensions the classifier weights through regularization or interactions that a univariate ranking cannot see. The model-agnostic correlation curve remained diffuse, with

many dimensions each weakly correlated with the outcome. The elasticnet logistic-regression curve was far more concentrated, the regularization having selected a compact predictive subset from a broadly informative space. Only $|\rho|$ is used, because the sign of a correlation on an unnamed latent dimension is not interpretable.

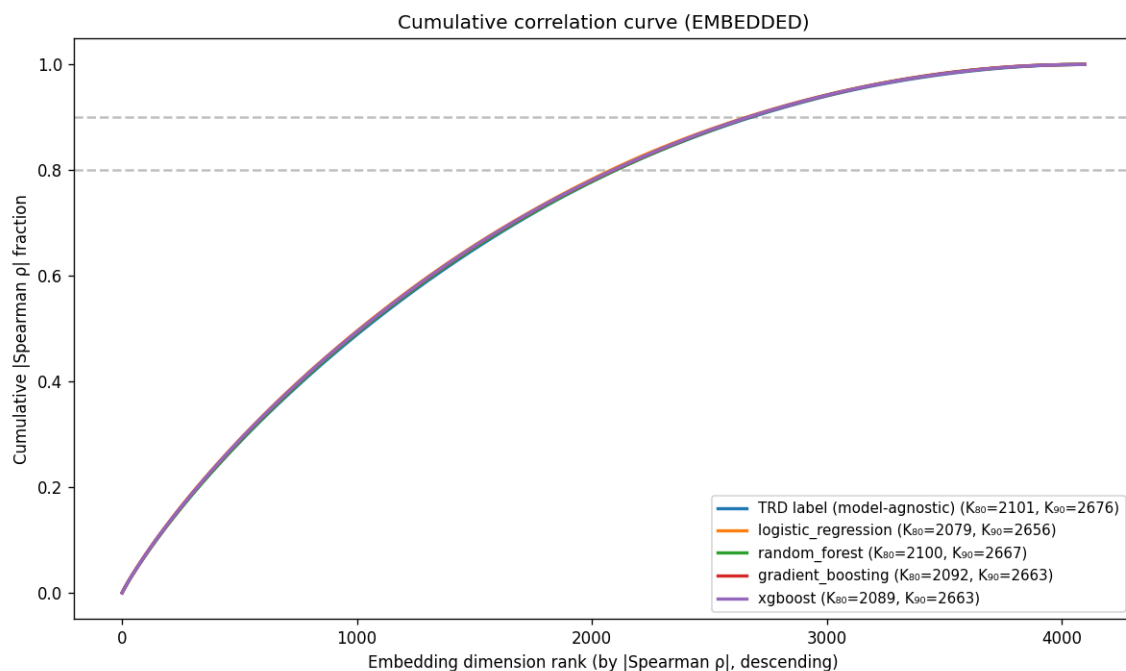
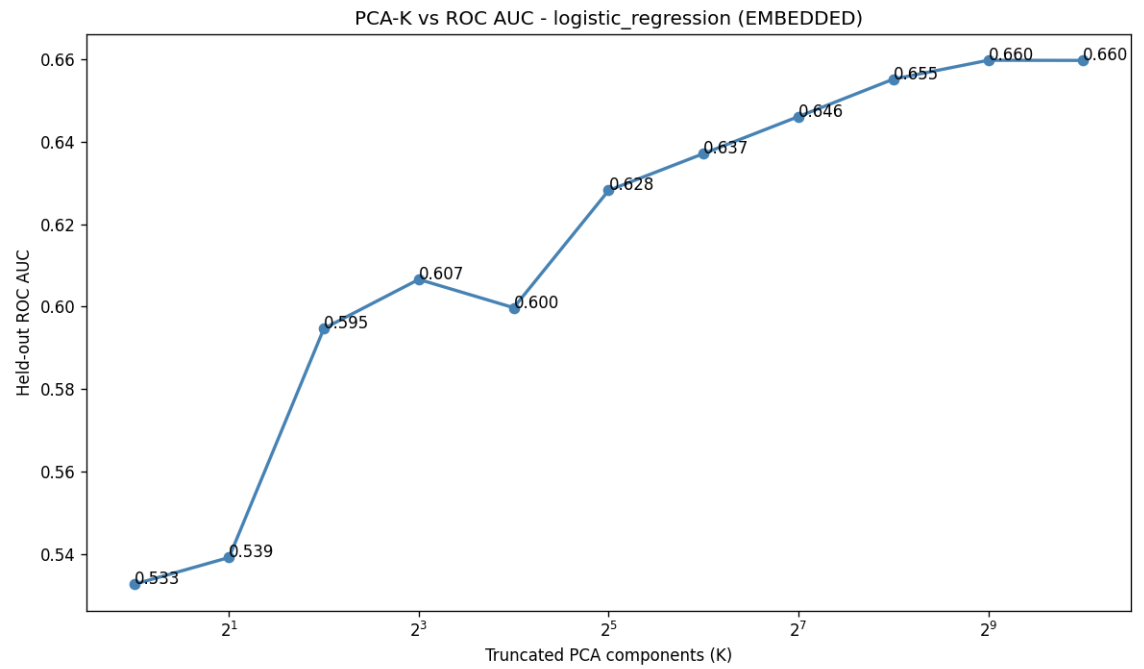


Figure S1. Cumulative absolute univariate (Spearman) correlation. One model-agnostic baseline curve ranks dimensions by $|\rho(\text{dim}, \text{outcome})|$. The four per-classifier curves rank by $|\rho(\text{dim}, \text{predicted risk})|$. Cumulative fraction of total $|\rho|$ mass versus rank, with K_{80} / K_{90} knees in the legend.

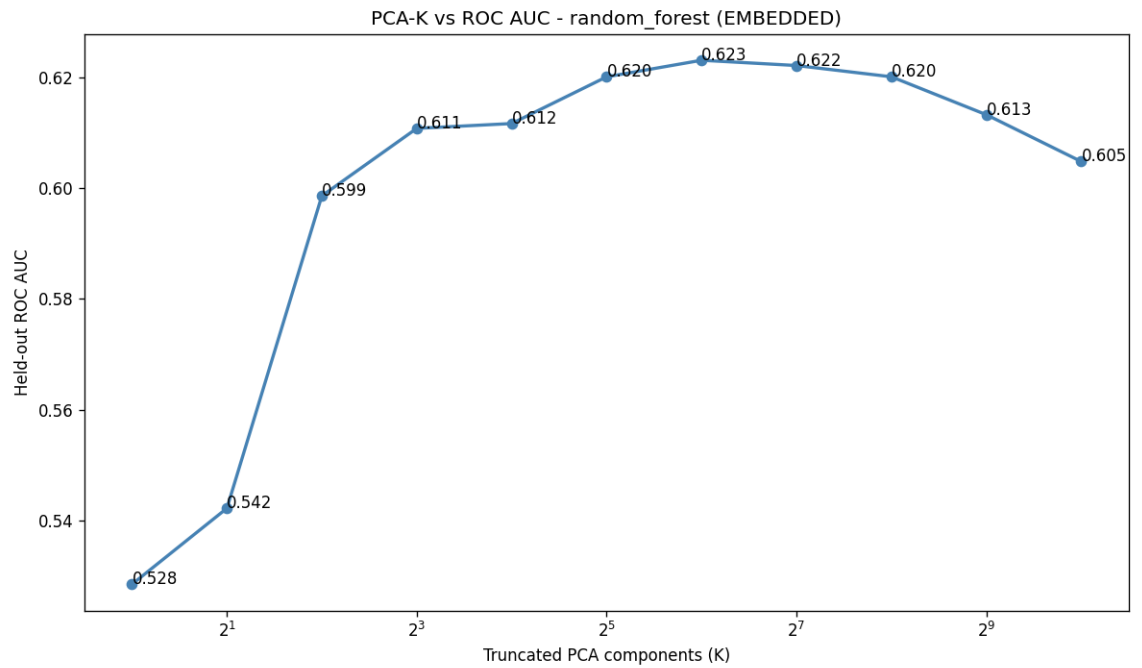
S1.3 PCA-K discrimination sweep

To locate the geometric plateau, each classifier was retrained on the top $K \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024\}$ principal components of the embedding and its held-out ROC AUC plotted against K (Figure S2). Discrimination rose steeply over the first handful of components and then plateaued, with no single low-dimensional projection recovering the full-rank performance. That matches the diffuse model-agnostic curve in S1.2. The compact subset in S1.1 is what the elasticnet penalty selects out of a broadly informative space, not a low-dimensional subspace the signal already occupies.

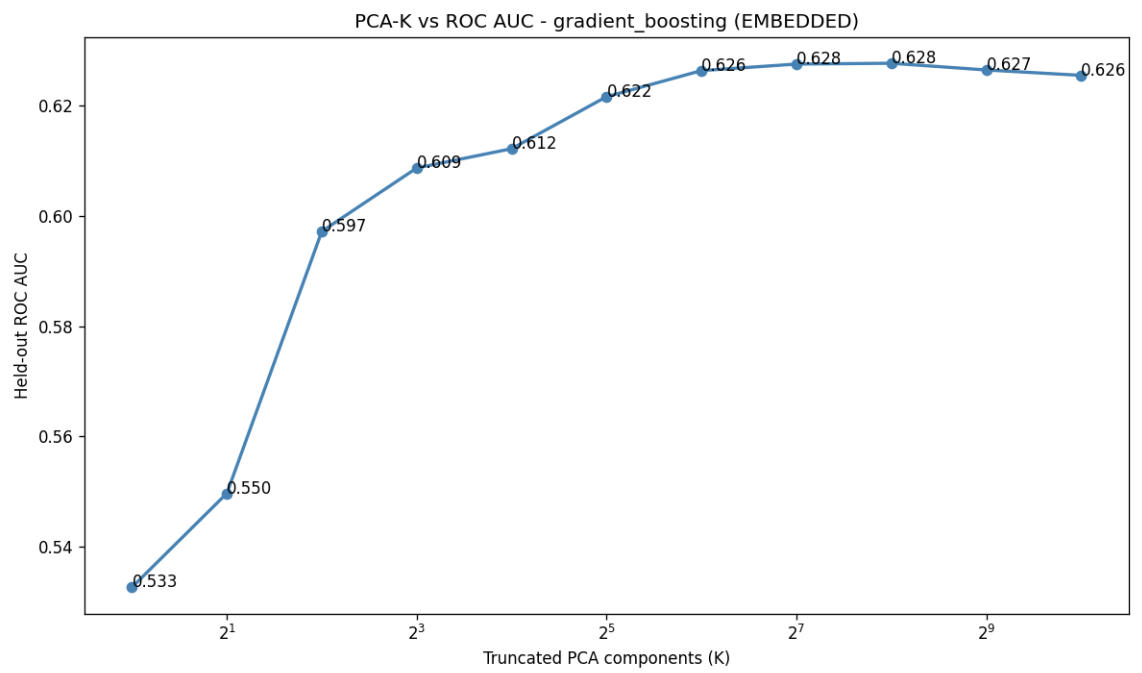
(A) Logistic regression



(B) Random forest



(C) Gradient boosting



(D) XGBoost

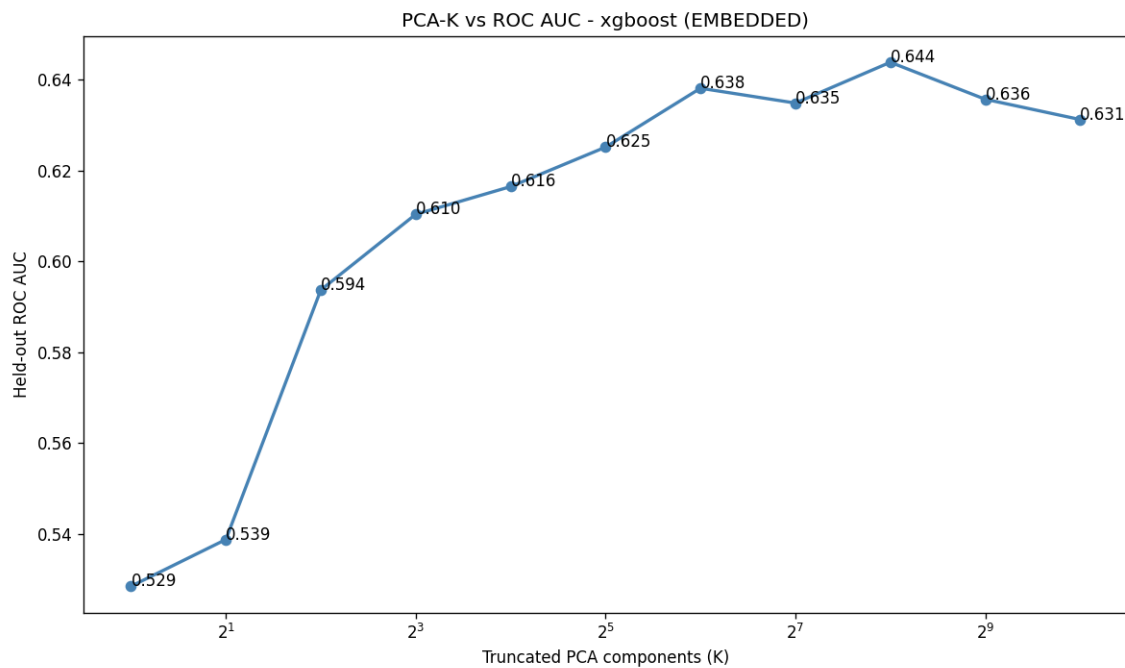


Figure S2. ROC AUC versus number of retained principal components, one panel per classifier: (A) logistic regression, (B) random forest, (C) gradient boosting, (D) XGBoost. The plateau marks the effective number of principal directions beyond which added components do not improve discrimination.

Supplement S2. Precision-recall performance

The main text reports discrimination as ROC AUC. Under the cohort's ~5:1 class imbalance (17.5% TRD-positive), ROC AUC can flatter apparent performance because the true-negative-dominated specificity term stays high regardless of how the positive class is ranked. The area under the precision-recall curve (AUPRC) is the complementary view: its no-skill baseline is the positive rate itself (0.175), not 0.5, and it exposes the precision cost of capturing TRD cases. We report it here rather than in the main text.

AUPRC tracked ROC AUC closely and remained modest for every model (Table S1): at this base rate, capturing high recall on the TRD

proxy necessarily forces low precision. No clinical role is inferred from these values. The main text’s Discussion, *Implications and future directions*, records that temporal and external validation are prerequisites for any such claim.

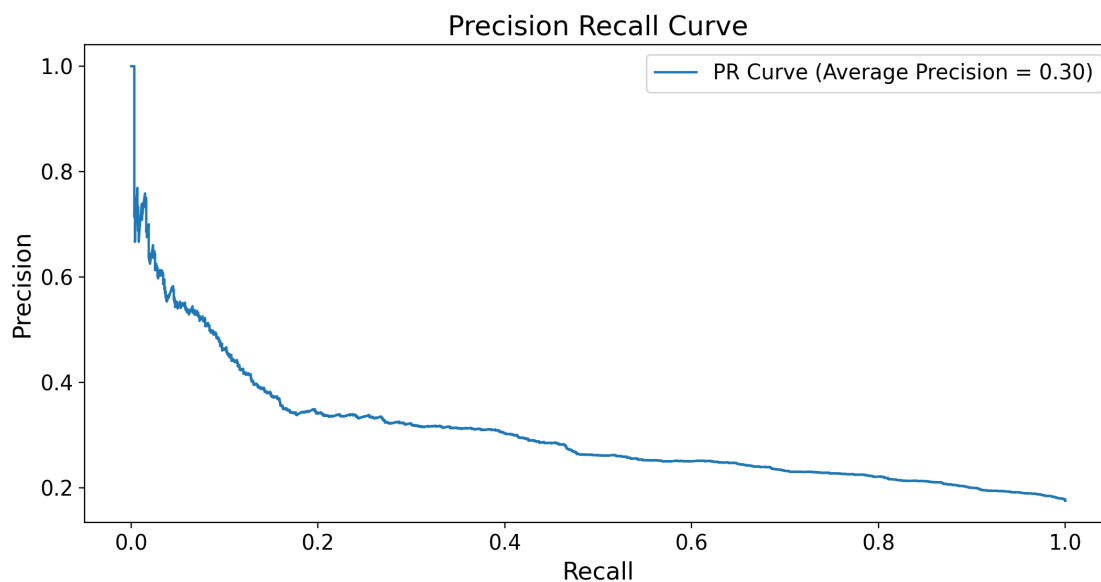
Table S1. AUPRC of the four classifiers on each representation (held-out test set). No-skill baseline = 0.175 (the positive rate).

Representation	Classifier	AUPRC
EMBEDDED	Logistic regression	0.302
EMBEDDED	Random forest	0.270
EMBEDDED	Gradient boosting	0.276
EMBEDDED	XGBoost	0.278
FEATURE	Logistic regression	0.278
FEATURE	Random forest	0.293
FEATURE	Gradient boosting	0.288
FEATURE	XGBoost	0.298

Table S2. Embedded logistic-regression AUPRC by encoder (held-out test set). No-skill baseline = 0.175.

Encoder	AUPRC
bge-small-en-v1.5	0.281
bge-en-icl	0.297
Qwen3-Embedding-4B	0.297
Qwen3-Embedding-8B	0.302

(A) Embedded logistic regression



(B) Feature-vector XGBoost

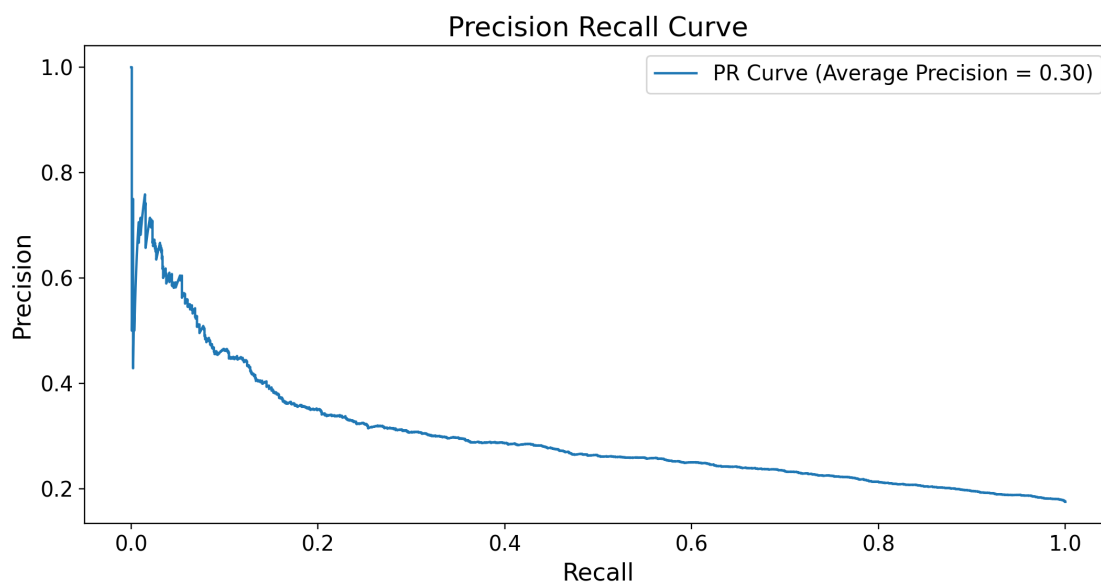


Figure S3. Precision-recall curves for the best classifier on each representation (held-out test set; primary Qwen3-Embedding-8B encoder). (A) Embedded logistic regression; (B) feature-vector XGBoost. The horizontal reference is the no-skill baseline (0.175, the positive

rate).

Supplement S3. Calibration absolute-error metrics

The main text reports calibration shape via the calibration slope and intercept (main-text Table 3) and the calibration curves (main-text Figure 5). Here we report the two absolute-error summaries: the Brier score and a weighted calibration error (WCE), both lower-is-better (Table S3). Brier scores were uniformly near 0.137–0.140 across all eight models, dominated by the base rate of 17.5% positive. The WCE separated the models more. Gradient boosting on the embedded representation was lowest (0.004) and logistic regression on the feature vector next (0.005), consistent with their near-ideal slopes in main-text Table 3.

Table S3. *Brier score and weighted calibration error (WCE) of the four classifiers on each representation (held-out test set). Lower is better for both. Calibration slope and intercept are in main-text Table 3.*

Representation	Classifier	Brier	WCE
EMBEDDED	Logistic regression	0.137	0.010
EMBEDDED	Random forest	0.140	0.006
EMBEDDED	Gradient boosting	0.139	0.004
EMBEDDED	XGBoost	0.139	0.013
FEATURE	Logistic regression	0.139	0.005
FEATURE	Random forest	0.139	0.018
FEATURE	Gradient boosting	0.138	0.010
FEATURE	XGBoost	0.138	0.012

Supplement S4. Example patient narratives

The EMBEDDED representation encodes a deterministically rendered Markdown narrative of each patient’s pre-index window (Methods). To make the encoder’s input concrete, we reproduce two representative narratives below, one TRD-positive and one TRD-negative. Both were extracted by `scripts/pipeline/neighbors/narrative_audit.py`, which takes the first patient by sorted de-identified hash in each

outcome class (output written to results/Qwen-Qwen3-Embedding-8B/google_medgemma-27b-text-it/narrative_audit/). Each narrative contains only structured EHR-derived fields, with no free text, names, or direct identifiers. The section headers and field order are fixed across all patients. Absent findings are rendered explicitly (“Absent”) and unrecorded values as “Missing”, so every patient is presented to the encoder through the same constant template.

The renderer’s own field labels predate the manuscript’s terminology: where the narrative prints MDD-to-anchor gap and Baseline window, the manuscript says the MDD-to-index gap and the lookback window. The narratives are reproduced exactly as the encoder received them and are not re-labeled here.

Three things visible in these examples are not matched by the feature vector. All three are recorded as a limitation of the head-to-head comparison (main text, Methods, *Predictors and patient representations*, and Supplement S8). The PHYSICAL HEALTH section renders the within-patient mean vital signs that were dropped from the feature vector at load time (main text, *Missing data and sample partition*). The SOCIODEMOGRAPHICS / ACCESS line renders a recorded sexual-orientation field that is absent from the feature inventory entirely, and it renders an unrecorded value there as the literal token nan rather than as Missing. The COHORT & INDEX line renders the index date, for which the feature vector has no column. The MEDICATION BURDEN and TREATMENT EXPOSURE sections also name individual ingredients where the feature vector carries only their counts, while pre-index history length is a feature-vector column that the narrative does not render.

TRD-positive example.

COHORT & INDEX

Condition: MDD (Dysthymia, Unspecified) | Index date: 2022-10-06 | Baseline window: -730...0 days | MDD-to-anchor gap: 57 days | Encounters in window: 34 | MDD within window: Present

SOCIODEMOGRAPHICS / ACCESS

Sex: Male | PreferredLanguage: English | AgeInYears: 37 | SexualOrientation: nan | M SD0H: None Recorded

PHYSICAL HEALTH

BMI: 29.8 | BP (mean): 140/88

PSYCH HISTORY

SOCIAL_ANXIETY: Absent | OCD: Absent | ANXIETY: Present | ADJUSTMENT_DISORDER: Absent
SUICIDE_FLAG (2y): Absent

SUBSTANCE ABUSE: Nicotine

MEDICAL COMORBIDITY

CHRONIC_PAIN: Absent | HYPERLIPIDEMIA: Absent | THYROID: Absent | DIABETES: Present

TREATMENT EXPOSURE

Prior adequate AD trials: MIRTAZAPINE: 0 | SSRI: 1 | VORTIOXETINE: 0 | BUPROPION: 0 |

Benzodiazepine days (2y): 0

Hypnotics: eszopiclone

Augmentation: Absent

MEDICATION BURDEN

Active meds at baseline: 0 (Absent)

NSAID burden: 0 (Absent)

UTILIZATION

Psych inpatient days: 0 (2y) | ED psych visits: 0 (2y)

SAFETY

UNCONTROLLED_HTN: Present | EPILEPSY: Absent

TRD-negative example.

COHORT & INDEX

Condition: MDD (Recurrent, Moderate) | Index date: 2025-05-09 | Baseline window: -730...0 days | MDD-to-anchor gap: 0 days | Encounters in window: 2 | MDD within window: Present

SOCIODEMOGRAPHICS / ACCESS

Sex: Female | PreferredLanguage: English | AgeInYears: 52 | SexualOrientation: nan | SDOH: None Recorded

PHYSICAL HEALTH

BMI: 35.6 | BP (mean): 138/86

PSYCH HISTORY

SOCIAL_ANXIETY: Absent | OCD: Absent | ANXIETY: Present | ADJUSTMENT_DISORDER: Absent

SUICIDE_FLAG (2y): Absent

SUBSTANCE ABUSE: Nicotine

MEDICAL COMORBIDITY

CHRONIC_PAIN: Absent | HYPERLIPIDEMIA: Absent | THYROID: Absent | DIABETES: Absent

TREATMENT EXPOSURE

Prior adequate AD trials: MIRTAZAPINE: 0 | SSRI: 0 | VORTIOXETINE: 0 | BUPROPION: 0 |

Benzodiazepine days (2y): 0

Hypnotics: Absent

Augmentation: Absent

MEDICATION BURDEN

Active meds at baseline: 0 (Absent)

NSAID burden: 0 (Absent)

UTILIZATION

Psych inpatient days: 0 (2y) | ED psych visits: 0 (2y)

SAFETY

UNCONTROLLED_HTN: Present | EPILEPSY: Absent

Supplement S5. Discrimination stratified by length of pre-index history

The main text tests the volume and recency confound by correlating the TRD label with three proxies for how much data a patient has, and finds all three associations weak and negative (Results, *Validity checks*; Figure 11). That test asks whether the *outcome* tracks data volume. This section asks the stronger question: whether *discrimination* tracks it, which is what one would expect if the predictor were partly reading data volume rather than clinical content.

The held-out test set was divided into quintiles of pre-index history length and the neighbor-weighted predictor was scored within each quintile. The nearest-retrieval, cosine-weighted arm is reported in Table S4. The pattern is the same under uniform weighting.

Table S4. Neighbor-weighted discrimination by quintile of pre-index history length (nearest retrieval, cosine weighting, embedded representation, held-out test set). Quintile bounds are in days of pre-index history.

Pre-index history (days)	Patients	ROC AUC
731-1,110	1,706	0.580
1,110-1,554	1,704	0.598
1,554-2,066	1,701	0.582
2,066-2,733	1,704	0.610
2,733-5,288	1,701	0.578

Discrimination is flat across the range. The spread between the best and worst quintile is 0.032 ROC AUC with no monotone trend, across a span of history length that varies more than fourfold from the shortest quintile to the longest. Had the predictor been exploiting record volume, discrimination should have risen with it. It does not.

Two limits on this analysis. It covers the neighbor-weighted arm only, not the trained classifiers that carry the main comparison, because the stratification was computed over the neighbor predictions. And each quintile holds roughly 1,700 patients, so the per-quintile estimates are individually imprecise. The claim rests on the absence of a trend rather than on any single value. A parallel stratification by embedding-neighborhood density was computed but is not reported, because the density distribution is concentrated enough that three of its five bins were empty, leaving too little coverage to interpret.

Supplement S6. The held-out set against the training set

The main text reports the split's sizes and outcome counts (Methods, *Missing data and sample partition*) and states that every per-predictor standardized mean difference (SMD) is small. This section shows the distributions behind that statement, because a claim about comparability that is asserted rather than displayed cannot be checked by a reader.

The split is the frozen stratified 80/20 split every evaluation in this paper uses. Nothing was refit or re-split to produce this table. SMDs use the same pooled-standard-deviation definition as main-text Table 1, so the two tables are directly comparable. Over the 100 predictor rows the largest absolute SMD is **0.036**, and no row reaches the conventional 0.1 threshold for a non-trivial imbalance. The maximum sits on a social-determinant flag recorded for 22 training patients and none of the held-out patients. That is a small-cell artifact rather than a distributional difference, and the flag's prevalence is 0.1% in the arm that has it. The outcome rate matches by construction, the split having been stratified on it, and is therefore not evidence of anything.

Table S5. Held-out patients against training patients on the characteristics reported in main-text Table 1. Continuous variables are median (IQR); categorical and boolean entries are n (%). SMD is training minus held-out, in pooled standard deviations. Vital signs are shown because they describe the cohort, but they are rendered only in the narrative representation and are absent from the feature vector as published (Supplement S8).

Characteristic	Training	Held-out	SMD
Demographics			
Age (years)	55 (38-70)	55 (38-70)	-0.014
Age band: 18-29	4,428 (13.0%)	1,085 (12.7%)	+0.008
Age band: 30-44	7,198 (21.1%)	1,827 (21.5%)	-0.008
Age band: 45-64	10,632 (31.2%)	2,636 (31.0%)	+0.006
Age band: 65+	11,805 (34.7%)	2,968 (34.9%)	-0.004
Sex: Female	24,682 (72.5%)	6,168 (72.4%)	+0.001
Race: White/Caucasian	27,244 (80.0%)	6,835 (80.3%)	-0.007
Race: Black/African American	1,894 (5.6%)	474 (5.6%)	-0.000
Race: Am. Indian/Alaska Native	1,637 (4.8%)	396 (4.7%)	+0.007
Race: Missing	247 (0.7%)	50 (0.6%)	+0.017
Language: English only	33,698 (98.9%)	8,425 (98.9%)	-0.000
Depression phenotype			
MDD recurrence: Recurrent	9,036 (26.5%)	2,177 (25.6%)	+0.022
MDD severity: Severe	1,818 (5.3%)	468 (5.5%)	-0.007
MDD severity: Moderate	5,807 (17.0%)	1,426 (16.7%)	+0.008
Psychiatric and substance comorbidity			
Suicidality flagged	1,392 (4.1%)	361 (4.2%)	-0.008
Anxiety disorder	17,975 (52.8%)	4,426 (52.0%)	+0.016
Substance use disorder (any)	7,472 (21.9%)	1,793 (21.1%)	+0.021
Insomnia	7,452 (21.9%)	1,898 (22.3%)	-0.010
PTSD	1,241 (3.6%)	300 (3.5%)	+0.006
Alcohol use disorder	1,523 (4.5%)	352 (4.1%)	+0.017
Medical comorbidity			
High cholesterol	12,591 (37.0%)	3,156 (37.1%)	-0.002
Uncontrolled hypertension	12,544 (36.8%)	3,136 (36.8%)	+0.000
Vital signs (narrative only)			
BMI	29 (25-35)	29 (25-35)	-0.013
Systolic BP	126 (118-136)	126 (118-136)	-0.010
BMI: not recorded	7,484 (22.0%)	1,911 (22.4%)	-0.011
Treatment and utilization			
Active med count	1 (0-2)	1 (0-2)	+0.008
Encounter count	21 (8-46)	20 (8-47)	-0.001
Pre-index history (days)	1792 (1213-2547)	1797 (1222-2541)	+0.004
Prior adequate AD trial (any class)	5,014 (14.7%)	1,262 (14.8%)	-0.003
Benzodiazepine days recorded	3,775 (11.1%)	949 (11.1%)	-0.002
Hypnotic recorded	1,319 (3.9%)	342 (4.0%)	-0.007
Augmentation therapy used	326 (1.0%)	96 (1.1%)	-0.017

What this does and does not establish. It establishes that no measured difference separates the patients the models were scored on from the patients they were trained on. The reported discrimination is therefore not an artifact of an evaluation sample that differs

on anything this table covers. It establishes nothing about representativeness of any wider population. The delivered extract is case-enriched by design (Methods, *Study design and data source*), and both halves of the split inherit that enrichment equally. A table comparing two halves of a selected cohort cannot detect the selection they share. The calibration reported in this paper remains calibration to this cohort's own base rate, and the limitation stands as written.

Supplement S7. Subgroup discrimination and calibration

This paper makes no fairness claim (Methods, *Model development, Semantic-feature ablation*, Discussion, *Limitations*). The permutation slate it reports measures how much a model's discrimination depends on a concept being present in its input, which is a different question from whether performance is even across groups. This section reports the subgroup evidence itself.

S7.1 Design

Nothing was refit. The per-patient held-out predicted probabilities the pipeline persists were partitioned by stratum, and discrimination and calibration were recomputed inside each group. Subsetting a per-patient prediction vector is arithmetically identical to scoring that model on the subset, so no published number can drift. Refitting within a subgroup would answer a different question, namely how a model trained only on one group predicts that group, and would break comparability with everything else reported.

All three prediction arms are covered. The embedded and feature-vector arms contribute their four classifiers each, and the neighbor-weighted arm contributes its two reported nearest-retrieval weightings, uniform and cosine. Only nearest-retrieval configurations are contrasted across groups. Whether a deliberately uninformative retrieval scheme is evenly uninformative across subgroups is not a fairness question, and including the negative controls would inflate the multiplicity burden on the contrasts that are.

Strata. Six sociodemographic families (sex, race, age band, marital status, smoking status, religion) and two clinical ones (MDD recurrence and severity). Race is collapsed to majority against recorded minority because a level-by-level breakdown is not estimable at 80.0% White/Caucasian. Preferred language is absent by necessity rather

than choice: 98.9% of the cohort prefers English, leaving one estimable level and therefore no contrast. A group whose smaller class holds fewer than 20 outcome events is declared not estimable rather than reported.

Intervals and multiplicity. Within-group intervals are percentile bootstraps over that group's own patients. Between-group intervals are *unpaired* bootstraps, because two subgroups are disjoint patient sets and each must be resampled independently, unlike the paired test used for the representation contrasts in main-text Figure 2. Two hundred forty contrasts were computed. At that number, nominally significant results arise from sampling alone, so each contrast carries a two-sided bootstrap *P* value taken from the same draws, adjusted across the entire reported set by Benjamini-Hochberg, controlling the false discovery rate at 5%. The adjusted value is what we interpret. The reported set is the 240 contrasts this paper describes, not the 288 the analysis ran. The two neighbor weightings that left the paper took 48 contrasts with them. A Benjamini-Hochberg threshold depends on how many tests it ranks, so dropping them changes every adjusted value.

Calibration slope is the coefficient of a logistic regression of the outcome on the logit of predicted risk (1.0 is perfect). Calibration-in-the-large is mean predicted risk minus observed rate (0.0 is perfect). Both are computed directly rather than from the binned calibration surface used in main-text Table 3, because a bin-mean fit over a few thousand patients describes the bin grid more than it describes the model.

S7.2 Sociodemographic strata

Table S6. *Discrimination and calibration by sociodemographic stratum, one representative model per arm, held-out test set. Groups are not disjoint across families: every patient with a recorded sex appears in one sex row and every patient with a recorded race in one race row.*

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
All held- out pa- tients	8,516	1,491	Embedded (lo- gis- tic re- gres- sion)	0.657 (0.643- 0.672)	0.137	0.97	+0.000
All held- out pa- tients	8,516	1,491	Feature vec- tor (lo- gis- tic re- gres- sion)	0.629 (0.613- 0.644)	0.139	0.95	-0.000
All held- out pa- tients	8,516	1,491	Neighbor weighted (near- est, co- sine)	0.594 (0.578- 0.610)	0.142	0.56	-0.007
Male	2,347	386	Embedded (lo- gis- tic re- gres- sion)	0.653 (0.622- 0.682)	0.131	0.94	-0.003
Male	2,347	386	Feature vec- tor (lo- gis- tic re- gres- sion)	0.626 (0.595- 0.655)	0.133	0.86	-0.003

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Male	2,347	386	Neighbor weighted (near- est, co- sine)	0.595 (0.563- 0.627)	0.135	0.58	-0.005
Female	6,168	1,105	Embedded (lo- gis- tic re- gres- sion)	0.658 (0.641- 0.675)	0.139	0.98	+0.002
Female	6,168	1,105	Feature vec- tor (lo- gis- tic re- gres- sion)	0.630 (0.612- 0.648)	0.141	0.99	+0.001
Female	6,168	1,105	Neighbor weighted (near- est, co- sine)	0.592 (0.574- 0.611)	0.145	0.55	-0.008
White/Caucasian	6,835	1,181	Embedded (lo- gis- tic re- gres- sion)	0.657 (0.640- 0.674)	0.136	0.97	+0.002

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
White/Caucasian	6,835	1,181	Feature vec- tor (lo- gis- tic re- gres- sion)	0.633 (0.615- 0.652)	0.137	0.98	+0.001
White/Caucasian	6,835	1,181	Neighbor weighted (near- est, co- sine)	0.604 (0.585- 0.622)	0.140	0.61	-0.007
Non- White (recorded)	1,631	302	Embedded (lo- gis- tic re- gres- sion)	0.652 (0.620- 0.684)	0.144	0.95	-0.004
Non- White (recorded)	1,631	302	Feature vec- tor (lo- gis- tic re- gres- sion)	0.608 (0.573- 0.643)	0.147	0.79	-0.004
Non- White (recorded)	1,631	302	Neighbor weighted (near- est, co- sine)	0.551 (0.515- 0.588)	0.152	0.37	-0.006

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Race not recorded	50	8	Embedded (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
Race not recorded	50	8	Feature vec- tor (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
Race not recorded	50	8	Neighbor- weighted (near- est, co- sine)	not es- timable	—	—	—
Age band: 18- 29	1,085	218	Embedded (lo- gis- tic re- gres- sion)	0.611 (0.568- 0.654)	0.156	0.78	+0.003
Age band: 18- 29	1,085	218	Feature vec- tor (lo- gis- tic re- gres- sion)	0.570 (0.524- 0.615)	0.160	0.60	+0.012

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Age band: 18- 29	1,085	218	Neighbor weights (near- est, co- sine)	0.546 (0.503- 0.589)	0.162	0.36	-0.004
Age band: 30- 44	1,827	372	Embedded (lo- gis- tic re- gres- sion)	0.653 (0.621- 0.682)	0.154	0.95	-0.002
Age band: 30- 44	1,827	372	Feature vec- tor (lo- gis- tic re- gres- sion)	0.636 (0.605- 0.667)	0.155	1.00	-0.002
Age band: 30- 44	1,827	372	Neighbor weights (near- est, co- sine)	0.598 (0.566- 0.629)	0.159	0.64	-0.017
Age band: 45- 64	2,636	473	Embedded (lo- gis- tic re- gres- sion)	0.652 (0.623- 0.679)	0.140	0.98	+0.007

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Age band: 45- 64	2,636	473	Feature vec- tor (lo- gis- tic re- gres- sion)	0.607 (0.580- 0.636)	0.143	0.88	-0.001
Age band: 45- 64	2,636	473	Neighbor weighted (near- est, co- sine)	0.592 (0.564- 0.622)	0.145	0.62	-0.005
Age band: 65+	2,968	428	Embedded (lo- gis- tic re- gres- sion)	0.658 (0.630- 0.682)	0.117	1.10	-0.005
Age band: 65+	2,968	428	Feature vec- tor (lo- gis- tic re- gres- sion)	0.641 (0.614- 0.669)	0.119	1.18	-0.003
Age band: 65+	2,968	428	Neighbor weighted (near- est, co- sine)	0.576 (0.546- 0.608)	0.122	0.40	-0.005

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Marital sta- tus: Di- vorced	882	211	Embedded (lo- gis- tic re- gres- sion)	0.640 (0.593- 0.681)	0.173	0.90	-0.046
Marital sta- tus: Di- vorced	882	211	Feature vec- tor (lo- gis- tic re- gres- sion)	0.607 (0.563- 0.649)	0.177	0.85	-0.049
Marital sta- tus: Di- vorced	882	211	Neighbor weighted (near- est, co- sine)	0.589 (0.542- 0.631)	0.181	0.39	-0.065
Marital sta- tus: Never Mar- ried	2,364	438	Embedded (lo- gis- tic re- gres- sion)	0.634 (0.609- 0.661)	0.145	0.85	+0.013
Marital sta- tus: Never Mar- ried	2,364	438	Feature vec- tor (lo- gis- tic re- gres- sion)	0.598 (0.569- 0.627)	0.149	0.72	+0.016

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Marital sta- tus: Never Mar- ried	2,364	438	Neighbor weighte (near- est, co- sine)	0.572 (0.543- 0.602)	0.151	0.47	+0.001
Marital sta- tus: Now Mar- ried	4,251	694	Embedde (lo- gis- tic re- gres- sion)	0.670 (0.650- 0.690)	0.129	1.05	-0.001
Marital sta- tus: Now Mar- ried	4,251	694	Feature vec- tor (lo- gis- tic re- gres- sion)	0.644 (0.623- 0.666)	0.131	1.10	-0.002
Marital sta- tus: Now Mar- ried	4,251	694	Neighbor weighte (near- est, co- sine)	0.606 (0.583- 0.631)	0.134	0.67	-0.002
Marital sta- tus: Sep- a- rated	99	21	Embedde (lo- gis- tic re- gres- sion)	0.518 (0.364- 0.673)	0.171	0.08	+0.001

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Marital sta- tus: Sep- a- rated	99	21	Feature vec- tor (lo- gis- tic re- gres- sion)	0.518 (0.357- 0.687)	0.169	0.30	-0.003
Marital sta- tus: Sep- a- rated	99	21	Neighbor weighted (near- est, co- sine)	0.565 (0.407- 0.723)	0.164	0.44	-0.007
Marital sta- tus: Wid- owed	891	120	Embedde (lo- gis- tic re- gres- sion)	0.654 (0.597- 0.708)	0.111	1.09	+0.022
Marital sta- tus: Wid- owed	891	120	Feature vec- tor (lo- gis- tic re- gres- sion)	0.625 (0.572- 0.678)	0.112	1.17	+0.017
Marital sta- tus: Wid- owed	891	120	Neighbor weighted (near- est, co- sine)	0.554 (0.492- 0.606)	0.117	0.33	+0.006

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Smoking sta- tus: Cur- rent Smoker	1,126	243	Embedded (lo- gis- tic re- gres- sion)	0.667 (0.631- 0.706)	0.158	1.01	-0.009
Smoking sta- tus: Cur- rent Smoker	1,126	243	Feature vec- tor (lo- gis- tic re- gres- sion)	0.617 (0.575- 0.660)	0.162	0.87	-0.011
Smoking sta- tus: Cur- rent Smoker	1,126	243	Neighbor weighted (near- est, co- sine)	0.613 (0.573- 0.652)	0.164	0.75	-0.020
Smoking sta- tus: For- mer Smoker	2,532	443	Embedded (lo- gis- tic re- gres- sion)	0.663 (0.636- 0.690)	0.137	0.98	+0.003
Smoking sta- tus: For- mer Smoker	2,532	443	Feature vec- tor (lo- gis- tic re- gres- sion)	0.633 (0.603- 0.660)	0.140	0.89	+0.001

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Smoking sta- tus: Former Smoker	2,532	443	Neighbor- weighted (near- est, co- sine)	0.578 (0.547- 0.607)	0.143	0.50	-0.010
Smoking sta- tus: Never Smoker	4,788	793	Embedded (lo- gis- tic re- gres- sion)	0.646 (0.623- 0.665)	0.132	0.93	+0.001
Smoking sta- tus: Never Smoker	4,788	793	Feature vec- tor (lo- gis- tic re- gres- sion)	0.625 (0.603- 0.645)	0.133	0.99	+0.002
Smoking sta- tus: Never Smoker	4,788	793	Neighbor- weighted (near- est, co- sine)	0.589 (0.566- 0.610)	0.136	0.51	-0.003
Religion: Catholic	608	93	Embedded (lo- gis- tic re- gres- sion)	0.695 (0.634- 0.754)	0.120	1.29	+0.009

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Religion: Catholic	608	93	Feature vec- tor (lo- gis- tic re- gres- sion)	0.648 (0.587- 0.710)	0.124	1.06	+0.002
Religion: Catholic	608	93	Neighbor weighted (near- est, co- sine)	0.611 (0.547- 0.676)	0.126	0.75	+0.009
Religion: Non- Christian	50	16	Embedding (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
Religion: Non- Christian	50	16	Feature vec- tor (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
Religion: Non- Christian	50	16	Neighbor weighted (near- est, co- sine)	not es- timable	—	—	—

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
Religion: Or- tho- dox	9	3	Embedded (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
Religion: Or- tho- dox	9	3	Feature vec- tor (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
Religion: Or- tho- dox	9	3	Neighbor- weighted (near- est, co- sine)	not es- timable	—	—	—
Religion: 759 Other/Unknown	145	145	Embedded (lo- gis- tic re- gres- sion)	0.615 (0.565– 0.664)	0.151	0.72	-0.013
Religion: 759 Other/Unknown	145	145	Feature vec- tor (lo- gis- tic re- gres- sion)	0.589 (0.537– 0.642)	0.153	0.66	-0.010

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration slope	Calibration- in-the- large
Religion: 759 Other/Unknown		145	Neighbor weights (near- est, co- sine)	0.549 (0.499– 0.602)	0.154	0.39	-0.016
Religion: 4,654 Protes- tant		793	Embed- ded (lo- gis- tic re- gres- sion)	0.665 (0.644– 0.685)	0.134	1.00	+0.003
Religion: 4,654 Protes- tant		793	Feature vec- tor (lo- gis- tic re- gres- sion)	0.642 (0.620– 0.662)	0.135	1.05	+0.002
Religion: 4,654 Protes- tant		793	Neighbor weights (near- est, co- sine)	0.599 (0.577– 0.622)	0.139	0.60	-0.007

S7.3 Clinical strata

Table S7. Discrimination and calibration by recorded depression phenotype, one representative model per arm.

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
MDD sever- ity: Mild	633	105	Embedded (lo- gis- tic re- gres- sion)	0.683 (0.624- 0.742)	0.130	1.39	-0.020
MDD sever- ity: Mild	633	105	Feature vec- tor (lo- gis- tic re- gres- sion)	0.607 (0.547- 0.667)	0.135	0.86	-0.018
MDD sever- ity: Mild	633	105	Neighbor weight (near- est, co- sine)	0.531 (0.475- 0.594)	0.140	0.28	-0.020
MDD sever- ity: Mod- er- ate	1,426	255	Embedded (lo- gis- tic re- gres- sion)	0.652 (0.618- 0.687)	0.142	0.98	+0.003
MDD sever- ity: Mod- er- ate	1,426	255	Feature vec- tor (lo- gis- tic re- gres- sion)	0.634 (0.599- 0.670)	0.142	1.06	+0.001

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
MDD sever- ity: Mod- er- ate	1,426	255	Neighbor weighted (near- est, co- sine)	0.587 (0.548- 0.624)	0.146	0.58	-0.003
MDD sever- ity: Psy- chotic	18	8	Embed- (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
MDD sever- ity: Psy- chotic	18	8	Feature vec- tor (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
MDD sever- ity: Psy- chotic	18	8	Neighbor weighted (near- est, co- sine)	not es- timable	—	—	—
MDD sever- ity: Re- mis- sion	143	19	Embed- (lo- gis- tic re- gres- sion)	not es- timable	—	—	—

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
MDD sever- ity: Re- mis- sion	143	19	Feature not es- vec- tor (lo- gis- tic re- gres- sion)	—	—	—	—
MDD sever- ity: Re- mis- sion	143	19	Neighbor not es- weighted timable (near- est, co- sine)	—	—	—	—
MDD sever- ity: Se- vere	468	137	Embedded 0.700 (lo- (0.648- gis- 0.749) tic re- gres- sion)	0.187	1.21	+0.029	
MDD sever- ity: Se- vere	468	137	Feature 0.636 vec- (0.578- tor 0.690) (lo- gis- tic re- gres- sion)	0.197	0.88	+0.029	
MDD sever- ity: Se- vere	468	137	Neighbor 0.665 weighted (0.611- (near- 0.718) est, co- sine)	0.194	1.10	+0.012	

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration slope	Calibration- in-the- large
MDD sever- ity: Un- spec- ified	5,828	967	Embedded (lo- gis- tic re- gres- sion)	0.641 (0.621- 0.660)	0.133	0.94	-0.000
MDD sever- ity: Un- spec- ified	5,828	967	Feature vec- tor (lo- gis- tic re- gres- sion)	0.615 (0.595- 0.634)	0.134	0.98	-0.001
MDD sever- ity: Un- spec- ified	5,828	967	Neighbor weight (near- est, co- sine)	0.578 (0.559- 0.598)	0.137	0.45	-0.008
MDD re- cur- rence: (un- recorded)	32	7	Embedded (lo- gis- tic re- gres- sion)	not es- timable	—	—	—
MDD re- cur- rence: (un- recorded)	32	7	Feature vec- tor (lo- gis- tic re- gres- sion)	not es- timable	—	—	—

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
MDD re- cur- rence: (un- recorded)	32	7	Neighborhood weighted (near- est, co- sine)	not es- timable	—	—	—
MDD re- cur- rence: Dys- thymia	584	97	Embedder (lo- gis- tic re- gres- sion)	0.690 (0.638– 0.748)	0.130	1.33	+0.008
MDD re- cur- rence: Dys- thymia	584	97	Feature vec- tor (lo- gis- tic re- gres- sion)	0.627 (0.565– 0.687)	0.134	1.13	+0.010
MDD re- cur- rence: Dys- thymia	584	97	Neighborhood weighted (near- est, co- sine)	0.592 (0.531– 0.655)	0.136	0.74	+0.002
MDD re- cur- rence: Re- cur- rent	2,177	433	Embedder (lo- gis- tic re- gres- sion)	0.698 (0.671– 0.726)	0.146	1.09	-0.000

Group	n	TRD+	Arm	ROC AUC (95% CI)	Brier	Calibration- slope	Calibration- in-the- large
MDD re- cur- rence: Re- cur- rent	2,177	433	Feature vec- tor (lo- gis- tic re- gres- sion)	0.676 (0.648- 0.703)	0.148	1.07	-0.002
MDD re- cur- rence: Re- cur- rent	2,177	433	Neighbor weighted (near- est, co- sine)	0.645 (0.616- 0.674)	0.152	0.82	-0.005
MDD re- cur- rence: Sin- gle Episode	5,723	954	Embedded (lo- gis- tic re- gres- sion)	0.633 (0.612- 0.652)	0.134	0.86	+0.000
MDD re- cur- rence: Sin- gle Episode	5,723	954	Feature vec- tor (lo- gis- tic re- gres- sion)	0.605 (0.584- 0.626)	0.136	0.83	-0.000
MDD re- cur- rence: Sin- gle Episode	5,723	954	Neighbor weighted (near- est, co- sine)	0.565 (0.545- 0.585)	0.139	0.36	-0.009

S7.4 What survives correction

Fifty-eight of the 240 contrasts exclude zero unadjusted, and **24 survive Benjamini-Hochberg adjustment**. They are not the ones the analysis was commissioned to look for.

Table S8. *Contrasts surviving Benjamini-Hochberg adjustment across all 240 reported comparisons, grouped by contrast and arm. “Models surviving” counts how many of that arm’s contrasted models cleared the threshold, out of four for the two classifier arms and two for the neighbor-weighted arm.*

Contrast	Arm	Models surviving	Δ ROC AUC range	smallest p (BH)
Age: 18-29 vs rest	Feature vector	1 of 4	-0.068	0.0400
Marital status: Never Married vs rest	Embedded	2 of 4	-0.054 to -0.051	0.0133
MDD recurrence: Recurrent vs rest	Embedded	4 of 4	+0.059 to +0.072	0.0133
MDD recurrence: Recurrent vs rest	Feature vector	4 of 4	+0.060 to +0.068	0.0133
MDD recurrence: Recurrent vs rest	Neighbor-weighted	2 of 2	+0.076	0.0133
MDD recurrence: Single Episode vs rest	Embedded	4 of 4	-0.071 to -0.065	0.0133
MDD recurrence: Single Episode vs rest	Feature vector	3 of 4	-0.064 to -0.056	0.0253

Contrast	Arm	Models surviving	Δ ROC AUC range	smallest p (BH)
MDD recurrence: Single Episode vs rest	Neighbor-weighted	2 of 2	-0.074 to -0.073	0.0133
MDD severity: Severe vs rest	Neighbor-weighted	2 of 2	+0.087	0.0400

Sex: no difference anywhere. All ten male-minus-female contrasts include zero, across every arm and model, and the largest absolute difference is 0.012 ROC AUC. Nothing here needs qualification.

Race: consistent in direction, not significant after correction. All ten White-minus-non-White contrasts are positive, from +0.005 to +0.053 ROC AUC, and five exclude zero unadjusted, including both neighbor-weighted configurations, which is a structurally different predictor reproducing the same sign. None survives adjustment, and the smallest adjusted P is .11. Calibration moves the same way in the feature-vector arm, where the slope among non-White patients is 0.79 against 0.98 among White patients, while the embedded arm holds at 0.95 against 0.97.

The correct reading is neither “no difference” nor “a gap of 0.04”. This cohort lacks the power to settle it. The minority stratum carries 302 outcome events against 1,181, so its intervals are roughly twice as wide, and a real difference of the size observed could as easily have gone undetected. What the data do show is a sign that is consistent across ten contrasts and three independent predictor families, which is not the shape sampling noise usually takes, and a calibration deficit concentrated in one representation. Both are reported for that reason, and neither is claimed as established.

What does survive is how the depression is coded. Twenty-one of the 24 surviving contrasts concern MDD recurrence and severity. Discrimination is higher among patients coded with recurrent MDD (+0.059 to +0.076 against the rest) and lower among those coded single-episode (-0.074 to -0.056), in every arm. Severity behaves the same way in the neighbor-weighted arm, where discrimination is higher among patients coded severe (+0.087). The pattern is coherent and is as much about documentation as about phenotype. The models discriminate better where the diagnosis is recorded specifically and worse where the coding is left unspecified, which is the ma-

jority of this cohort. Of the 8,516 held-out patients, 5,828 are coded unspecified severity.

Two sociodemographic contrasts survive, and both point the same way as the documentation story. Discrimination is lower among patients aged 18-29 than among the rest in the feature-vector arm (-0.068 under XGBoost, adjusted $P = .04$). It is lower among never-married patients in the embedded arm, where two of the four classifiers survive, at -0.054 and -0.051 (adjusted $P = .01$). Younger and never-married patients accumulate less recorded history, which is consistent with thinner records supporting weaker prediction, though this analysis cannot separate that explanation from any other.

Subgroup discrimination, one representative model per arm

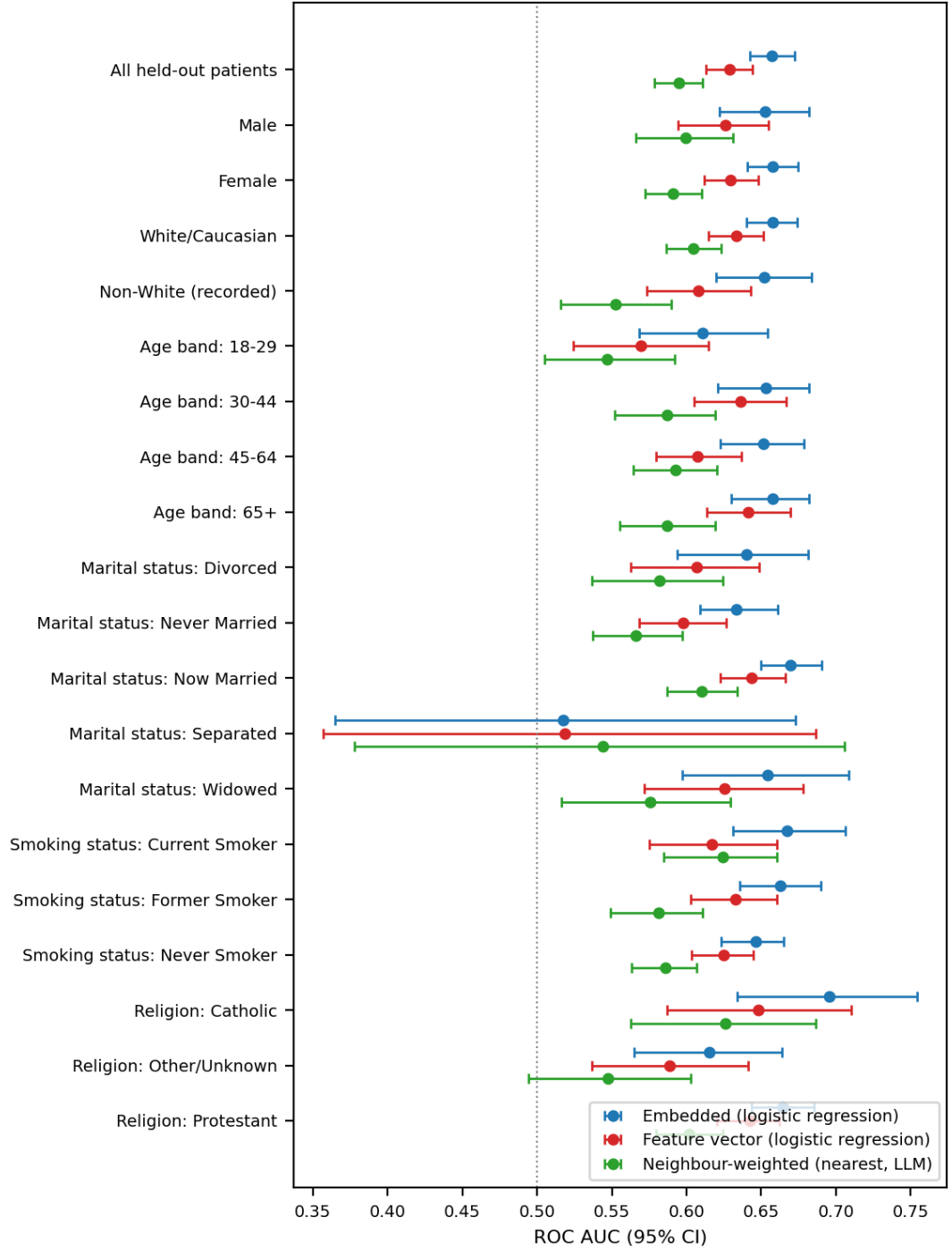


Figure S4. Subgroup discrimination across sociodemographic strata, one representative model per arm, with 95% bootstrap confidence intervals. The dotted line marks chance. Strata declared not estimable are omitted.

Supplement S8. Field-level crosswalk of the two representations

The head-to-head comparison in this paper is between two encodings of one record, so it is only a comparison of encodings to the extent that both encodings carry the same information. This section is the field-by-field accounting, and it does two jobs. **It bounds the null**, because the two representations do not render an identical inventory: 11 of the 28 rows below are asymmetric, and in 10 of them the narrative receives information the feature vector does not. **And it is the evidence for a scope condition the main text states in its own voice** (Methods, *Predictors and patient representations*; Discussion, *Principal findings*). Every row here traces to a field that predictor selection chose before either representation existed. The patient data were hand-picked in both arms, and neither was handed a raw record. A reader who takes the null to mean that the embedding found structure without being told where to look can check that against this table, row by row, and the answer is in the second column.

Table S9. Field-level crosswalk. **Asymmetric** marks a row where the two representations do not receive the same information. Missing-value rules are as implemented, not as intended.

Source field	Feature-vector encoding	Narrative rendering	Missing-value rule	Match
MDD recurrence, severity	two categorical columns	Condition: MDD (recurrence, severity)	empty string is a level in both	matched
Index date	no column	Index date: YYYY-MM-DD	never missing	asymmetric
Lookback window width	no column (constant)	Baseline window: -730...0 days	never missing	matched (carries no information)

Source field	Feature-vector encoding	Narrative rendering	Missing-value rule	Match
MDD-to-index gap	mdd_to_anchor float	MDDays, anchor gap: N days	never missing	matched
Encounter count in window	num_encounters float	Encounters in window: N	never missing	matched
MDD inside lookback window	mdd_within_window boolean	MDD, within window: Present/Absent	never missing	matched
Recorded pre-index history length	pre_anchor_history float	history_days, not rendered	never missing	asymmetric
Age	AgeInYears, float	AgeInYears: N	never missing	matched
Sex	categorical	Sex: X	absent value becomes its own one-hot level; narrative prints the raw token	matched
Race / ethnicity	categorical, seven levels	Race_Ethnicity: X	own one-hot level; narrative prints raw token	matched
Preferred language	categorical, collapsed to five levels	raw Epic value	own one-hot level	asymmetric (finer in narrative)
Marital status	categorical, collapsed to five levels	raw Epic value	own one-hot level	asymmetric (finer in narrative)

Source field	Feature-vector encoding	Narrative rendering	Missing-value rule	Match
Religion	categorical, collapsed to five levels	raw Epic value	own one-hot level	asymmetric (finer in narrative)
Smoking status	categorical, collapsed to three levels	raw Epic value	own one-hot level	asymmetric (finer in narrative)
Sexual orientation	no column	SexualOrientation X	rendered as the literal token nan when recorded-but-empty absence is an explicit False in the matrix, an omission from the list in the narrative	asymmetric
Social determinants of health	nine boolean columns	SDOH: list of present categories only	dropped, so no rule; narrative prints Missing never missing	matched in content
BMI, systolic BP, diastolic BP	dropped at load	BMI: N \ BP (mean) : S/D, or Missing ARM: Present/Absent for all eight	dropped, so no rule; narrative prints Missing never missing	asymmetric
Psychiatric comorbidity (8 flags)	eight boolean columns	SUICIDE FLAG (2y) : Present/Absent	never missing	matched
Suicidality flag	boolean	names of present substances only	absence explicit in matrix, omitted in narrative	matched
Substance-use specifics (10)	ten boolean columns			matched in content

Source field	Feature-vector encoding	Narrative rendering	Missing-value rule	Match
General medical comorbidity (4)	four boolean columns	ARM: Present/Absent for all four	never missing	matched
Prior adequate AD trials	five trials_<class> counts	Prior adequate AD trials: CLASS: n for all five	never missing	matched
Benzodiazepine days	benzo_days_count	Benzodiazepine days (2y): N	never missing	matched
Hypnotics	hypnotics_burden count only	current plus ingredient names	never missing	asymmetric (finer in narrative)
Active medications	polypharmacy count only	count plus ingredient names	never missing	asymmetric (finer in narrative)
NSAIDs	nsaid_count, count only	count plus ingredient names	never missing	asymmetric (finer in narrative)
Psychiatric inpatient days, ED visits	two float columns	Psych inpatient days: N \ ED psych visits: N	never missing	matched
Prescribing-safety comorbidity (2)	two boolean columns	ARM: Present/Absent	never missing	matched

What the crosswalk shows. The imbalance is not symmetric, and it is larger than a count of rows suggests. Eleven rows are asymmetric, and in ten of them the narrative receives information the feature vector does not. Two are fields it renders and the feature vector has no column for, the index date and sexual orientation. One is the row of vital signs dropped at load. Four are sociodemographic fields where the feature vector collapses raw Epic values into coarse categories and the narrative prints the raw token. Three are medication blocks where the feature vector carries a count and the narrative names the ingre-

dients. The eleventh runs the other way: recorded history length, which the narrative did not render.

Two rules are worth stating because they are easy to assume wrongly. Missingness is *not* silently imputed on the categorical side of the feature vector: an absent value becomes its own one-hot level, so the model can use the fact of a missing record. And the vital signs were dropped rather than imputed, so the published feature arm never saw them in any form. The reason is mechanical rather than principled: a numeric column carrying nulls cannot pass through the pipeline's scaler.

The main text states the imbalance as a bound on the comparison (Methods, *Predictors and patient representations*). This table is what makes that bound checkable, row by row.

Supplement S9. Subgroup outcome prevalence

This section reports the *within-subgroup prevalence* of TRD, meaning the number of TRD-positive patients divided by the total patients in that subgroup. Main-text Table 1 gives column percentages conditioned on TRD status, not row percentages, so the prevalences below were computed from the raw subgroup counts. For instance, the TRD prevalence for the 18-29 age band is 20.6% (1,135 TRD-positive patients out of 5,513).

TRD prevalence varied modestly across subgroups. By sex it was 18.0% (female) versus 16.2% (male). By age band it followed a clear gradient: the two youngest bands were highest and nearly tied, at 20.6% in 18-29-year-olds (1,135/5,513) and 20.3% in 30-44-year-olds (1,836/9,025). Prevalence then fell through 18.4% at 45-64 (2,445/13,268) to 13.8% in those aged 65 or older (2,039/14,773). Several small social-determinant strata showed elevated but imprecise prevalence (upbringing-related issue 33.0%, legal/criminal issue 26.1%, employment issue 24.0%, family/support-group issue 22.9%). Patients with missing race/ethnicity ($n = 297$) did not show elevated prevalence (15.5% against the 17.5% base rate). Strata this small are why the subgroup performance analysis in Supplement S7 declares some groups not estimable. One example is a psychosocial-circumstances flag, with $n = 12$.